Excel Model Walkthrough

A worked example: how the budget model answers the headline question, sheet by sheet. FinTech Customer Acquisition and Profitability Analysis. Synthetic data, extract date 2026-09-30, all amounts in EUR.

Concept in brief

Spending more on a channel wins more customers, but each extra euro wins fewer than the one before (diminishing returns). A budget model describes that curve for each channel, puts a value on a customer, and then finds the spend per channel that leaves the most profit. Excel does the last part with Solver. This document follows the workbook from its inputs to its answer.

Problem

The bank spent 28,603 EUR on paid channels in the last quarter (July to September 2026): paid social 19,030, paid search 5,375, affiliate 3,116 and referral 1,081. How should next quarter's budget be reallocated between the four channels? Each channel must keep at least 50% and may get at most 150% of its baseline spend, referral at most 120%, and the total may not exceed the baseline budget.

Given / Find

Given	Find
Results of three SQL views: the channel scorecard (CAC, 12-month value, credit loss), the monthly spend and customers per channel (24 months), and the cumulative margin curve. Inputs: KYC cost 2.50 EUR per check, limits 50% / 150% / 120%, credit loss multiplier 1.0.	The spend per channel that maximises net profit, the profit it gives against the baseline spend, and how stable that answer is when the inputs change.

Strategy

Work through the sheets in the order the numbers flow. Each sheet takes the one before it as input, so a change in one cell reaches the answer without any copying:

Order	Sheet	What it holds
1	Data_Scorecard, Data_Monthly, Data_Curve	The SQL view results, pasted as values
2	Assumptions	Every input and scenario lever, one cell each
3	Unit_Economics	CAC, LTV, payback, ROAS, risk-adjusted value per customer
4	Curves	Customers = a x spend^b for each channel, fitted on the data
5	Optimizer	The decision cells, the limits, the profit, and the Solver settings
6	Scenarios	The best allocation under eight different sets of assumptions
7	Summary and Notes	The answer on one page, and the method and limits

Solution, step by step

STEP 1	The data sheets: what comes from SQL
What	Three SQL views are pasted as values: **rpt.v_channel_scorecard** (5 rows), **rpt.v_budget_inputs** (96 rows: 4 channels x 24 months) and **rpt.v_channel_curve** (60 rows: 5 channels x 12 months). The only formulas on these sheets are the natural logs on Data_Monthly (columns F and G), used to fit the curves, and a flag on Data_Curve that marks the first month a channel reaches its CAC.
Why	Every metric is defined once, in SQL. Excel does not recalculate CAC or credit loss from raw tables; it uses the SQL results and adds only what SQL cannot: editable assumptions, curves and optimization.
Result	The scorecard shows what each channel spent, how many customers funded, and the 12-month value per customer. Organic has no ad spend, so it has no CAC.

acquisition_channel	ad_spend	funded_customers	kyc_decisions	kyc_cost	cac	ad_only_cac	eligible_customers_12m	ltv_12m	revenue_12m	ltv_cac
affiliate	23677.59	846	1368	3420	32.03	27.99	382	251.54	321.21	7.853
organic	0	1401	1831	4577.5			650	167.97	217.02	
paid_search	45681.2	1545	2086	5215	32.94	29.57	719	185.36	237.31	5.627
paid_social	73973.35	1225	2550	6375	65.59	60.39	562	171.28	216.32	2.611
referral	8718.79	911	1056	2640	12.47	9.57	467	165.36	217.06	13.262

Data_Scorecard (first 11 of 17 columns). Source: SQL view rpt.v_channel_scorecard.

channel_name	month_start	ad_spend	credited_customers	in_baseline_quarter	ln_spend	ln_customers
affiliate	Oct 2024	645.39	18.10	0	6.4699	2.8959
affiliate	Nov 2024	1,351.87	36.40	0	7.2092	3.5946
affiliate	Dec 2024	1,702.98	42.80	0	7.4401	3.7565
affiliate	Jan 2025	760.88	39.75	0	6.6345	3.6826
affiliate	Feb 2025	816.17	18.38	0	6.7046	2.9113
affiliate	Mar 2025	642.03	38.13	0	6.4646	3.6410
affiliate	Apr 2025	874.68	26.87	0	6.7739	3.2910
affiliate	May 2025	676.64	38.77	0	6.5171	3.6576
affiliate	Jun 2025	868.20	25.10	0	6.7664	3.2229
affiliate	Jul 2025	697.36	21.43	0	6.5473	3.0648
affiliate	Aug 2025	819.03	25.40	0	6.7081	3.2347
affiliate	Sep 2025	935.72	38.40	0	6.8413	3.6481
affiliate	Oct 2025	1,067.93	35.93	0	6.9735	3.5816

Data_Monthly (first rows). Spend and customers per channel and month, with the logs used for the fit.

STEP 2 Assumptions: every input in one place

What	Each input has its own cell on the Assumptions sheet (blue on yellow): the KYC cost, the months in a quarter, the two scenario levers (budget multiplier and credit loss multiplier), the three limits, a switch for the Summer Reach months, and the interest earned after month 12.
Why	The formulas never contain a typed number. Changing a scenario means changing one cell, and the whole workbook follows.
Result	Two inputs matter most: the credit loss multiplier (1.0 = observed losses) and the Summer Reach switch (1 = leave the wasteful campaign out of the paid social curve).

Assumptions

Blue cells on yellow are inputs. Everything else in the workbook is calculated from them and from the SQL data sheets.

Input	Value	Unit	Where it comes from / what it does
KYC cost per identity check	2.50	EUR	assumptions table (kyc_cost_per_check). Part of CAC.
Months in the planning quarter	3	months	Converts the monthly response curve into a quarterly number.
Budget multiplier (scenario lever)	1.00	x	1.0 = the baseline budget (July to September 2026 spend). 0.8 = minus 20 percent.
Credit loss multiplier (scenario lever)	1.00	x	1.0 = observed credit losses. 1.25 = default losses 25 percent higher.
Lower limit per channel	50.0%	% of baseline	Project decision: each channel keeps at least 50 percent of its baseline spend.
Upper limit per channel	150.0%	% of baseline	Project decision: each channel gets at most 150 percent of its baseline spend.
Upper limit for referral	120.0%	% of baseline	Project decision: referral volume is limited by the existing customer base.
Leave the SO_Summer_Reach months (June to September 2026) out of the paid social curve	1	1 = yes, 0 = no	The wasteful campaign ran in those months and distorts the curve. 1 fits paid social on the first 20 months.
Weak fit threshold (R squared)	0.30	R squared	Curves below this are flagged as weak fits.

Interest earned after month 12 (not in the 12-month LTV)

Channel	Value per customer (EUR)	Unit	Note
affiliate	-	EUR per customer	Default 0, as defined in the project. Measured on the 12-month cohort: 69.10 EUR per customer. Enter it to count interest after month 12.
organic	-	EUR per customer	Default 0, as defined in the project. Measured on the 12-month cohort: 43.51 EUR per customer. Enter it to count interest after month 12.
paid_search	-	EUR per customer	Default 0, as defined in the project. Measured on the 12-month cohort: 53.16 EUR per customer. Enter it to count interest after month 12.
paid_social	-	EUR per customer	Default 0, as defined in the project. Measured on the 12-month cohort: 48.08 EUR per customer. Enter it to count interest after month 12.
referral	-	EUR per customer	Default 0, as defined in the project. Measured on the 12-month cohort: 43.60 EUR per customer. Enter it to count interest after month 12.

Fixed project definitions (not inputs): CAC = (ad spend + KYC cost) / funded customers; LTV is the observed 12-month margin; risk-adjusted LTV = LTV minus credit loss per customer; the response curve is customers = a x spend^b.

Assumptions sheet.

STEP 3	Unit economics: what a customer costs and earns
What	Each channel row recomputes CAC = (ad spend + KYC decisions x KYC cost) / funded customers, LTV / CAC, ROAS and the payback month, then subtracts the credit loss to get the risk-adjusted value per customer, and the KYC cost per customer to get the **net value per customer**.
Why	The optimizer needs one number per channel: what an extra customer is worth after everything it costs and everything it can lose. That number is the net value per customer.
Result	Before credit loss every paid channel pays back within 12 months. After credit loss only referral keeps a positive value.

Channel	Has ad spend (1/0)	Ad spend (EUR)	Funded customers	KYC decisions	KYC cost (EUR)	CAC (EUR)	Ad-only CAC (EUR)	Customers in 12-month cohort	LTV 12 months, before credit loss (EUR)	Revenue 12 months (EUR)	LTV / CAC
affiliate	1	23,678	846	1,368	3,420	32.03	27.99	382	251.54	321.21	7.9x
organic	0	-	1,401	1,831	4,578	n/a	n/a	650	167.97	217.02	n/a
paid_search	1	45,681	1,545	2,086	5,215	32.94	29.57	719	185.36	237.31	5.6x
paid_social	1	73,973	1,225	2,550	6,375	65.59	60.39	562	171.28	216.32	2.6x
referral	1	8,719	911	1,056	2,640	12.47	9.57	467	165.36	217.06	13.3x

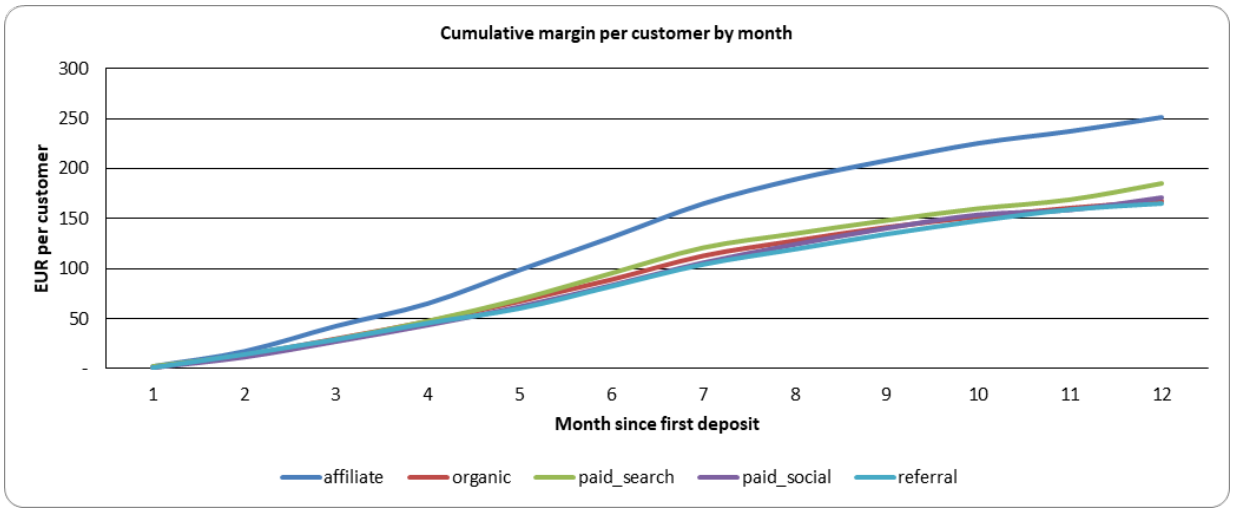
Channel	ROAS	Payback month	Credit loss per customer (EUR)	Interest after month 12 (EUR)	Risk-adjusted LTV (EUR)	Risk-adjusted LTV / CAC	KYC cost per funded customer (EUR)	Net value per customer (EUR)
affiliate	11.5x	3	742.49	-	(490.95)	-15.3x	4.04	(494.99)
organic	n/a	n/a	196.21	-	(28.24)	n/a	3.27	(31.51)
paid_search	8.0x	4	217.32	-	(31.96)	-1.0x	3.38	(35.34)
paid_social	3.6x	6	225.31	-	(54.03)	-0.8x	5.20	(59.23)
referral	22.7x	2	110.87	-	54.49	4.4x	2.90	51.59

Unit_Economics. Left: acquisition cost and 12-month value. Right: payback, credit loss, risk-adjusted value and net value per customer.

Worked check for affiliate

Step	Calculation	Result
CAC	(23,677.59 ad spend + 1,368 KYC decisions x 2.50) / 846 funded customers	32.03
LTV / CAC	251.54 / 32.03	7.9x
Payback	Cumulative margin per customer: month 2 = 17.53 (below 32.03), month 3 = 42.82 (above)	month 3
Risk-adjusted LTV	251.54 LTV + 0 interest after month 12 - 742.49 credit loss per customer	-490.95
Net value per customer	-490.95 - 4.04 KYC cost per funded customer (3,420 / 846)	-494.99

Payback: cumulative margin per customer by month



Each line is the average cumulative margin of the customers who have had that many full months. A channel pays back in the first month its line passes its CAC (referral 12.47 in month 2, affiliate 32.03 in month 3, paid search 32.94 in month 4, paid social 65.59 in month 6).

STEP 4	Curves: how many customers a euro buys
What	For each channel the sheet fits customers per month = a x monthly spend^b. It takes the logs of both sides, so the curve becomes a straight line, ln(customers) = ln(a) + b x ln(spend), and **LINEST** finds the slope b and the intercept. R squared says how well monthly spend explains monthly customers.
Why	The optimizer has to know what happens to the number of customers when the spend moves away from last quarter's level. A tiny example: with a = 2 and b = 0.5, a spend of 25 wins 10 customers, 100 wins 20 and 400 wins 40. Quadrupling the spend only doubles the customers.
Result	b is below 1 in every channel (0.41 to 0.69), so every channel has diminishing returns. The fits are rough: R squared runs from 0.25 (referral, flagged as a weak fit) to 0.70 (paid search).

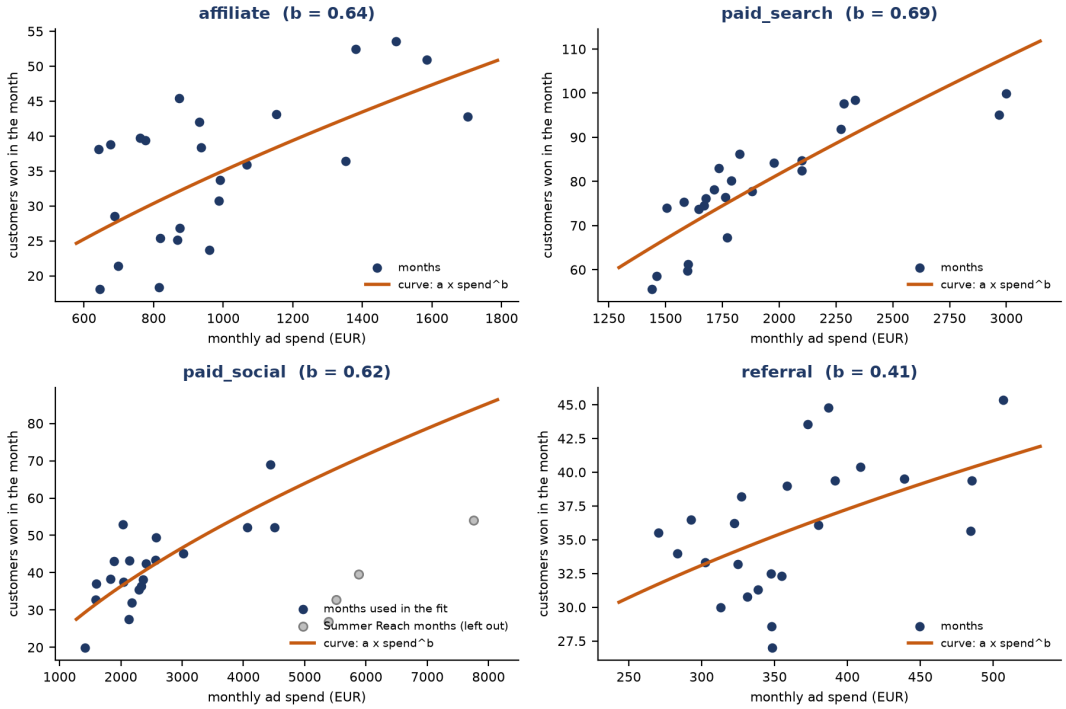
Response curves: customers = a x spend^b

Fitted on the log of monthly spend and monthly credited customers, 24 monthly points per channel (20 for paid social when the Summer Reach months are left out).

Channel	Points used	b (slope)	ln(a) (intercept)	a	R squared	Fit quality	Baseline monthly spend (EUR)	Curve: customers per month at baseline spend	Actual average customers per month, Jul to Sep 2026
affiliate	24	0.6407	-0.8698	0.4190	0.342	ok	1,039	35.9	38.5
paid_search	24	0.6913	-0.8518	0.4266	0.700	ok	1,792	75.7	74.0
paid_social	20	0.6163	-1.0916	0.3357	0.546	ok	6,343	74.0	40.1
referral	24	0.4114	1.1530	3.1676	0.251	weak fit	360	35.7	36.4

How to read it: b below 1 means diminishing returns (doubling the spend gives less than double the customers). A low R squared means monthly spend explains little of the monthly customers, so treat that curve as rough. Compare the last two columns: if the curve gives far more customers than were actually won, the baseline spend was inefficient (this is the case for paid social).

Curves sheet. The last two columns are a sanity check: customers the curve gives at the baseline spend, against customers actually won in July to September 2026. They agree for affiliate, paid search and referral. For paid social the curve gives 74 a month against 40 won, which shows that the baseline spend was inefficient.



Monthly spend against customers won, with the fitted curve. Paid social is fitted without the four Summer Reach months (grey points): that campaign spent heavily and won few customers, and would otherwise flatten the whole curve to b = 0.22.

STEP 5 Optimizer: letting Solver choose the spend

What	The yellow cells (column E) are the spend per channel, which Solver changes. For each channel the sheet computes the customers at that spend (months x a x (spend / months)^b), multiplies by the net value per customer, and adds up: **net profit = value of the customers - ad spend**.
Why	Solver searches for the four spends that make net profit as large as possible while the limits hold. The settings are stored in the sheet.
Result	Affiliate, paid search and paid social go to their lower limit (50% of baseline). Referral goes to its upper limit (+20%). Total spend falls from 28,603 to 15,058, and net profit improves from -97,564 to -56,843, an improvement of 40,720 per quarter.

Channel	Baseline spend, Jul to Sep 2026 (EUR)	Lower limit (EUR)	Upper limit (EUR)	Recommended spend (EUR) <- Solver changes these	Change vs baseline
affiliate	3,116	1,558	4,675	1,558	-50.0%
paid_search	5,375	2,687	8,062	2,687	-50.0%
paid_social	19,030	9,515	28,545	9,515	-50.0%
referral	1,081	541	1,298	1,298	20.0%
Total	**28,603**			**15,058**	**-47.4%**

Channel	Customers at baseline spend	Customers at recommended spend	Net value per customer (EUR)	Value of customers, baseline (EUR)	Value of customers, recommended (EUR)
affiliate	107.7	69.1	(494.99)	(53,309)	(34,191)
paid_search	227.1	140.7	(35.34)	(8,026)	(4,970)
paid_social	222.0	144.8	(59.23)	(13,152)	(8,580)
referral	107.1	115.4	51.59	5,526	5,956
Total	**664.0**	**470.0**		**(68,961)**	**(41,785)**

Budget ceiling (EUR)	28,603
Total recommended spend (EUR)	15,058
Net profit at baseline spend (EUR)	(97,564)
Net profit at recommended spend (EUR) <- Solver maximises this	**(56,843)**
Uplift: recommended minus baseline (EUR)	**40,720**
Spend within the budget ceiling?	OK
All channels inside their limits?	OK

Optimizer sheet after Solver has run. Top: spend limits and the recommended spend. Middle: customers and value at the baseline and at the recommended spend. Bottom: profit and the two checks.

How to run Solver

Setting	Value
Set Objective	$B$14 (net profit at the recommended spend)
To	Max
By Changing Variable Cells	$E$5:$E$8
Constraints	$E$5:$E$8 >= $C$5:$C$8 (lower limits); $E$5:$E$8 <= $D$5:$D$8 (upper limits); $E$9 <= $B$11 (budget ceiling)
Solving Method	GRG Nonlinear

Data > Solver, check the settings, press Solve, keep the Solver solution. If an assumption has changed, press Solve again.

Why the answer sits on the limits

Look at what one extra euro does in each channel. In paid search, an extra euro wins about 0.03 customers, each worth -35 EUR after credit losses, so the euro is lost and the customers lose more. In referral, an extra euro wins about 0.04 customers, each worth +52 EUR, so it brings back about 2.1 EUR for every 1 EUR spent. So the model spends as little as allowed where a customer loses money, and as much as allowed in referral. The limits decide the answer, not the curves.

Does the result depend on where Solver starts?

The problem is not smooth everywhere, so Solver was run from three different starting points: the baseline spend, the lower limits and the upper limits. All three ended on the same allocation (1,558 / 2,687 / 9,515 / 1,298) and the same profit (-56,843). An independent grid search outside Excel gave the same numbers.

STEP 6	Scenarios: how stable is the answer?
What	Eight scenarios were solved and pasted as values: the budget 20% lower and higher, credit losses 25% higher and lower, the paid social curve fitted on all 24 months, interest after month 12 counted, and the whole budget forced to be spent.
Why	A recommendation that flips when one assumption moves a little is not worth much.
Result	In every scenario with a budget ceiling the three channels stay at their lower limit. Referral stays at or near its upper limit; it falls slightly below its baseline only when credit losses rise 25%.

Scenario	Lever	Budget ceiling (EUR)	affiliate spend (EUR)	paid_search spend (EUR)	paid_social spend (EUR)	referral spend (EUR)	Total spend (EUR)
Base case	-	28,603	1,558	2,687	9,515	1,298	15,058
Budget -20%	Budget multiplier 0.8	22,882	1,558	2,687	9,515	1,298	15,058
Budget +20%	Budget multiplier 1.2	34,323	1,558	2,687	9,515	1,298	15,058
Credit losses +25%	Credit loss multiplier 1.25	28,603	1,558	2,687	9,515	1,032	14,792
Credit losses -25%	Credit loss multiplier 0.75	28,603	1,558	2,687	9,515	1,298	15,058
Paid social curve fitted on all 24 months	Summer Reach switch = 0	28,603	1,558	2,687	9,515	1,298	15,058
Interest after month 12 counted	Interest after month 12 = measured values	28,603	1,558	2,687	9,515	1,298	15,058
Budget fully spent (equality rule)	Constraint E9 = B11 instead of <=	28,603	1,558	2,687	22,564	1,298	28,107

Scenario	Customers won	Net profit at recommended spend (EUR)	Net profit at baseline spend (EUR)	Uplift (EUR)
Base case	470	(56,843)	(97,564)	40,720
Budget -20%	470	(56,843)	(97,564)	40,720
Budget +20%	470	(56,843)	(97,564)	40,720
Credit losses +25%	460	(88,648)	(145,370)	56,722
Credit losses -25%	470	(25,021)	(49,757)	24,736
Paid social curve fitted on all 24 months	448	(55,514)	(92,832)	37,318
Interest after month 12 counted	470	(32,595)	(62,702)	30,107
Budget fully spent (equality rule)	572	(75,921)	(97,564)	21,643

Scenarios sheet. The last row forces the whole budget to be spent: the model then moves the money into paid social, the channel that wins the fewest customers per euro, which only looks better because it loses less per euro. This is why the budget is a ceiling in the model.

STEP 7	Summary: the answer on one page
What	The Summary sheet links to the Optimizer and Unit_Economics sheets and shows the recommended allocation, the result and five conclusions.
Why	Someone who opens the workbook for a minute should see the answer before any formula.
Result	The recommended spend is also shown rounded to 100 EUR (1,600 / 2,700 / 9,500 / 1,300), which is the form a budget would take.

Budget model: which channels deserve next quarter's marketing budget?

FinTech Customer Acquisition and Profitability Analysis. Synthetic data, extract date 2026-09-30. All numbers in EUR.

Headline question

Which acquisition channels bring profitable customers, and how should next quarter's budget be reallocated?

Recommended allocation (Solver result, base case)

Channel	Baseline spend (EUR)	Recommended spend (EUR)	Change	Recommended, rounded to 100 (EUR)	Risk-adjusted LTV per customer (EUR)	CAC (EUR)
affiliate	3,116	1,558	-50.0%	1,600	(490.95)	32.03
paid_search	5,375	2,687	-50.0%	2,700	(31.96)	32.94
paid_social	19,030	9,515	-50.0%	9,500	(54.03)	65.59
referral	1,081	1,298	20.0%	1,300	54.49	12.47
Total	**28,603**	**15,058**	**-47.4%**	**15,100**		

Result

Net profit at baseline spend (EUR per quarter)	(97,564)
Net profit at recommended spend (EUR per quarter)	(56,843)
Improvement (EUR per quarter)	**40,720**

Summary sheet (top part). Below the allocation, the sheet lists five conclusions and how to read the colours.

Answer

Move the money out of affiliate, paid search and paid social, which lose money on every customer once credit losses are counted, and keep as much as allowed in referral. At the limits: affiliate 1,558 (-50%), paid search 2,687 (-50%), paid social 9,515 (-50%), referral 1,298 (+20%), total 15,058 against 28,603. Net profit improves by about 40,700 per quarter, but it stays negative (about -56,800), because loan losses dominate. The biggest lever is credit policy, not channel mix.

Check

- All 614 formulas in the workbook calculate without errors.
- CAC, LTV / CAC, payback and risk-adjusted LTV on Unit_Economics reproduce the SQL scorecard exactly (for example affiliate: CAC 32.03, LTV / CAC 7.9, risk-adjusted LTV -490.95).
- Solver was run in Excel from three starting points and always ended on the same allocation. A separate optimization outside Excel gave the same result to the cent.
- The result was tested against a different assumption (credit losses halved, below). Excel and the independent calculation agreed again.

Key takeaways

- Diminishing returns is a property of the curve (b below 1), and it is what makes "all the money into the best channel" a bad answer.
- The optimizer follows the value of a customer. When a customer loses money, no curve can justify more spend, so the limits decide the allocation.
- A model is only as good as its inputs. The curves are rough, the credit loss is observed on early customers, and the data is synthetic. Read the allocation as a direction, not as a forecast.
- Keep the assumptions in cells and the formulas free of typed numbers, so any scenario is one change away.

Try it yourself

On the Assumptions sheet, set the credit loss multiplier (B8) to 0.5, which halves the credit losses. Go to the Optimizer sheet, run Solver again, and look at the allocation. Before you do, guess: which channels will now be worth more than their lower limit?

Answer to the exercise

At half the credit losses, paid search (+73 EUR per customer) and paid social (+53) become profitable and affiliate is still negative (-124). Paid search goes to its upper limit (8,062) because each extra euro there returns more than a euro. Paid social stays at its lower limit (9,515): a customer is worth 53 EUR, but an extra euro wins so few customers (about 0.009) that it returns only about 0.50 EUR. The full result is: affiliate 1,558, paid search 8,062, paid social 9,515, referral 1,298, total 20,433. Net profit is +13,156 against -1,951 at the baseline spend, an improvement of 15,107. Set B8 back to 1.0 and run Solver again to return to the base case.

Limits of the model

- The data is synthetic and the default behaviour is an assumption, so the credit results are the most uncertain part.
- Credit loss covers the whole life of a loan, while the value per customer covers 12 months. Interest earned after month 12 (about 44 to 69 EUR per customer) is not counted by default; the Assumptions sheet has an input to test it.
- Loans issued after 2026-03-31 are too young to default, so the loss is a floor.
- Each curve rests on 24 monthly points, and the channels are treated as independent.
- The uplift is a model estimate. It has not been tested, and the recommendation is a hypothesis.